Phase-2 Solar Canopy and Energy Storage Feasibility Study

Document ID: DOC_ENG_004 | **Type:** FEASIBILITY_STUDY | **Publication:** 2024-11-10 | **Institution:** VESIT (VESIT Green Campus, Chembur)

FEASIBILITY STUDY: PHASE-2 SOLAR CANOPY AND ENERGY STORAGE SYSTEM

Document Reference: VESIT/FS/SOLAR-BESS/2024/02

Project Code: INIT_SOLAR_PH2

Commissioned by: Campus Decarbonization Taskforce and Governing Board.

Technical Consultants: Department of Electronics & Telecommunication (DEP_EXTC) and Department of Computer Engineering (DEP_CMPN).

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. EXECUTIVE RATIONALE AND DECARBONISATION TARGET ALIGNMENT

To sustain compliance with the institutional Net-Zero Roadmap (TRG_NET_ZERO) by 2040 and achieve the mandatory milestone of 50.0% renewable electricity penetration by 2030 (TRG_RENEWABLE), Vivekananda Education Society's Institute of Technology (VESIT Green Campus) must aggressively expand clean energy generation beyond existing terrace allocations. The 220 kWp Phase-1 installation (INIT_SOLAR_PH1) currently generates 312,400 kWh/year, satisfying 13.22% of gross campus electricity demand (2,362,400 kWh). However, building terraces across Academic Block A (BLD_ACA), Academic Block B (BLD_ACB), Administration Block (BLD_ADM), and Library (BLD_LIB) have reached maximum structural weight allowances.

This engineering study evaluates the technical and financial feasibility of Project INIT_SOLAR_PH2, proposing a 200 kWp solar PV expansion combined with a 100 kWh Battery Energy Storage System (BESS) deployed across ground parking surfaces and the Auditorium roof (BLD_AUD), advancing SDG 7 (Affordable and Clean Energy), SDG 9 (Industry, Innovation and Infrastructure), and SDG 13 (Climate Action).

2. SYSTEM ARCHITECTURE AND CIVIL-STRUCTURAL DESIGN

The proposed 200 kWp addition consists of two interconnected subsystems:

2.1 Central Parking Elevated Solar Canopy (150 kWp DC):

Installation of cantilevered structural steel canopies over the central two-wheeler and four-wheeler paved parking lots covering 1,850 m2 of surface area. The canopy will support 375 high-efficiency bifacial monocrystalline silicon modules (400 Wp each). Bifacial architecture captures direct downward solar irradiance as well as ground-reflected albedo from light-colored concrete pavers, yielding an estimated 8.5% bifacial boost. The canopy provides covered, shaded parking for 240 two-wheelers and 45 four-wheelers, integrating two Level-2 22 kW EV fast chargers for campus feeder transit (INIT_ELECTRIC_SHUTTLE).

2.2 Auditorium Perimeter Rooftop Array (50 kWp DC):

Placement of 125 bifacial 400 Wp PV modules on custom aluminum ballasted racking over the curved perimeter roof of the Auditorium (BLD_AUD, 2,800 m2), connecting directly to the BLD_AUD electrical distribution sub-board.

2.3 Battery Energy Storage System (100 kWh BESS):

A central containerized lithium iron phosphate (LiFePO4) battery system with 100 kWh nominal storage capacity and 50 kW continuous bi-directional inverter capability. LiFePO4 chemistry guarantees thermal stability in Mumbai's sub-tropical climate, delivering over 4,500 charge cycles at 80% depth of discharge.

3. ENERGY YIELD, GRID DISPLACEMENT, AND EMISSION ABATEMENT

PVSyst hydrological and solar simulation modeling calibrated with Mumbai solar irradiance datasets predicts an annual yield of 284,000 kWh for the 200 kWp Phase-2 arrays:

- Canopy Array (150 kWp): Generates 213,000 kWh/year.

- Auditorium Array (50 kWp): Generates 71,000 kWh/year.

When combined with Phase 1 (312,400 kWh), total institutional solar generation will reach 596,400 kWh/year, meeting 25.25% of total campus electrical consumption. Applying the CEA grid emission factor EF_ELEC_GRID (0.82 kg CO2e/kWh), the Phase-2 system will prevent 232,880 kg CO2e of Scope 2 emissions annually. The combined solar installations will avoid

489,048 kg CO2e every year.

4. DIESEL GENERATOR ABATEMENT AND PEAK LOAD SHAVING

The integration of the 100 kWh BESS provides high-impact operational decarbonisation:

- Peak Demand Management: Discharges 45 kWh daily during high-tariff evening hours (18:00 to 21:00) during postgraduate sessions, reducing contractual peak demand penalties by INR 2,40,000 annually.

- Emergency Backup & Diesel Abatement: The BESS offers sub-20ms uninterruptible transfer capability, shielding mission-critical server rooms in the Computer Laboratories (BLD_LAB) during instantaneous grid dropouts. This capability eliminates 45 hours of idling run-time on campus diesel generators, preventing the combustion of 2,400 litres of diesel fuel annually. Applying factor EF_DIESEL_GEN (2.68 kg CO2e/L), this eliminates 6,432 kg CO2e of direct Scope 1 emissions every year.

5. CAPITAL EXPENDITURE AND FINANCIAL APPRAISAL

- Total Turnkey Capital Expenditure (EPC): INR 1,48,00,000 (1.48 Crores), covering solar arrays, parking steelworks, BESS container, and grid tie-in.

- Annual Energy Savings: INR 31,24,000 (from 284,000 kWh at INR 11.00/kWh).

- Peak Demand and Diesel Fuel Savings: INR 4,81,600 (diesel savings of 2,400 L at INR 100.67/L + demand tariff cuts).

- Annual Net Operational Cashflow: INR 33,55,600 (net of O&M).

- Simple Payback Period: 4.41 years; Internal Rate of Return (IRR): 19.8% over a 25-year asset lifecycle.

The study conclusively advises immediate Board authorization and tendering.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_ENG_004**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*